

ECE 315: Signals and Systems

Homework 7

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This homework is due on **4/13, 11:59pm**. For analytical problems, write your solutions independently by hand. For coding problems, attach your source code and answers. Combine them in one PDF document and submit on CANVAS. Please ensure that your PDF is clearly readable on a computer.

1. (2pts: Parseval's theorem) Prove the Parseval's theorem for discrete-time periodic signals. I.e., show the following equality for a discrete-time periodic signal $x[n]$ with period N .

$$\frac{1}{N} \sum_{n=\langle N \rangle} |x[n]|^2 = \sum_{k=\langle N \rangle} |a_k|^2$$

where a_k is the Fourier series coefficients of $x[n]$.

2. (2pts: Fourier series representation, and its properties) Assume that we observe the following properties for a continuous-time periodic signal $x(t)$:

- $x(t)$ is periodic with period 3;
- $x(t)$ has Fourier series coefficients a_k that satisfy: (i) $a_k = a_{k+2}$; (ii) $a_k = a_{-k}$;
- $\int_{-1/2}^{1/2} x(t) dt = 1$;
- $\int_1^2 x(t) dt = 2$.

Determine the signal $x(t)$.

3. (2pts: Periodic convolution, Fourier series properties) Consider the periodic convolution of two periodic discrete-time signals $x[n]$ and $y[n]$ with common period N :

$$z[n] = \sum_{r=\langle N \rangle} x[r]y[n-r]$$

- (i) Show that $z[n]$ is periodic with period N ;
 - (ii) Assume that the Fourier series coefficients of $x[n]$ and $y[n]$ are a_k and b_k , respectively. Compute the Fourier series coefficients of $z[n]$ in terms of a_k and b_k ;
 - (iii) Assume $x[n] = \sin(\frac{3\pi n}{4})$ and $y[n] = \begin{cases} 1 & 0 \leq n \leq 1 \\ 0 & 2 \leq n \leq 7 \end{cases}$, and their common period $N = 8$. Compute the Fourier series representation of $z[n]$.
4. (2pts: Continuous-time LTI system response to periodic signals) Consider a continuous-time linear-time invariant system with impulse response $h(t) = e^{-t}u(t)$, and an input,

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\pi t}$$

where $w_0 = \pi$ is the fundamental frequency, and $a_k = 0$ except that

$$a_0 = 0, a_1 = a_{-1} = \frac{1}{4}, a_2 = a_{-2} = \frac{1}{2}, a_3 = a_{-3} = \frac{1}{3}.$$

- (i) Determine the system function (system's eigenvalue) $H(s)$ and its frequency response $H(jw)$;
 - (ii) Compute the output of the LTI system for the input $x(t)$.
5. (2pts: Discrete-time LTI system response to periodic signals) Consider a discrete-time linear-time invariant system with impulse response $h[n] = \frac{1}{2^n}u[n]$, and an input,
- $$x[n] = \frac{1}{2} + \cos(\frac{2\pi}{5}n) + \cos(\frac{4\pi}{5}n).$$
- (i) Determine the system function (system's eigenvalue) $H(z)$ and its frequency response $H(e^{jw})$;
 - (ii) Compute the output of the LTI system for the input $x[n]$.
6. Read Section **9.7-3**, and complete the problem **9.7-3** in **LSS**.